

Figure 7.22: Dark field microscopy of *Treponema pallidium*

- *Treponema pallidum* strains (Nichol's strain) are maintained in rabbit testes.

7.12.3 Pathogenesis

Source of infection – Human beings (patients)

Mode of transmission – Sexual contact

Site of entry – Through minute abrasions/ cuts on skin or mucosa

Incubation period – 10–90 days

- *Treponema pallidum* causes venereal syphilis, which is acquired by sexual contact. The pathogen enters the human body through cut on the skin or mucosa of genital areas.
- The clinical disease sets in after an incubation period of about a month. There are 3 clinical stage of venereal syphilis, namely – primary, secondary and tertiary syphilis.

Primary syphilis

- A papule appears on the genital area that ulcerates, forming a chancre of primary syphilis called hard chancre.
- The chancre is covered by thick exudates, very rich in spirochetes.
- The regional lymph nodes are swollen, discrete, rubbery and non – tender.
- Even before the chancre appears, the spirochetes spread from the site of entry into the lymph and bloodstream, so the patient may be infectious during the late incubation period.
- The chancre invariably heals within 10–40 days, even without treatment, leaving a thin scar.

Secondary syphilis

- Secondary syphilis sets in 1–3 months after the primary lesion heals. During this interval, the patient is asymptomatic.
- The secondary lesions are due to widespread multiplication of the spirochetes and dissemination through the blood.
- Secondary syphilis is characterized by appearance of papular skin rashes, mucous patches in the oropharynx and condylomata (a raised growth on the skin resembling a wart).
- The lesions are abundant in spirochetes and the patient is most infectious during the secondary stage.
- There may be retinitis (inflammation of the retina of the eye), meningitis, periostitis, and arthritis.
- Secondary lesions usually undergo spontaneous healing, in some cases taking as long as 4 or 5 years.
- After the secondary lesions disappear, there is a period of dormant known as **latent syphilis** the patient does not show any clinical symptoms but with positive serology.

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